PRAKTIKUM BASIS DATA

MODUL 09

Scripting Procedure and Function

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

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Jurnal 09

1. create table employees as select * from hremployees. select * from employees

   Jawaban:

Analisisi :

create table employees as select * from hremployees Perintah ini digunakan untuk membuat tabel baru bernama employees dengan cara menyalin seluruh struktur dan data dari tabel hremployees.dan juga select * from employees ini digunakan untuk menampilkan seluruh data yang ada pada tabel employees.

Query & Output :

```
create table employees as select * from hremployees;

select * from employees;
```

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.003 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90
4	103	Alexander	Hunold	AHUNOLD	590.423.4567	03-JAN-06	IT_PROG	9000	(null)	102	60
5	104	Bruce	Ernst	BERNST	590.423.4568	21-MAY-07	IT_PROG	6000	(null)	103	60
6	107	Diana	Lorentz	DLORENTZ	590.423.5567	07-FEB-07	IT_PROG	4200	(null)	103	60
7	124	Kevin	Mourgos	KMOURGOS	650.123.5234	16-NOV-07	ST_MAN	5800	(null)	100	50
8	141	Trenna	Rajs	TRAJS	650.121.8009	17-OCT-03	ST_CLERK	3500	(null)	124	50
9	149	Eleni	Zlotkey	EZLOTKEY	011.44.1344.429018	29-JAN-08	SA_MAN	10500	0.2	100	80
10	174	Ellen	Abel	EABEL	011.44.1644.429267	11-MAY-04	SA_REP	11000	0.3	149	80

2.

```oracle_plsql
CREATE OR REPLACE PROCEDURE increase_salary (
department_id_in   IN employees.department_id%TYPE,
    increase_pct_in    IN NUMBER)
     IS
BEGIN
    FOR employee_rec
            IN (SELECT employee_id
            FROM employees
           WHERE department_id = increase_salary.department_id_in)
    LOOP
            UPDATE employees emp
                SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pct_in
            WHERE emp.employee_id = employee_rec.employee_id;
            DBMS_OUTPUT.PUT_LINE ('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END increase_salary;
```

BEGIN increase_salary (90, .50); ROLLBACK; END; / select * from employees where department_id = 90

Jawaban:

Analisis :

Procedure increase_salary dibuat untuk menaikkan gaji karyawan berdasarkan:

- department_id
- persentase kenaikan gaji

Procedure ini menggunakan perulangan (FOR LOOP) untuk:

- Mengambil setiap employee_id pada department tertentu
- Mengupdate gaji masing-masing karyawan

BEGIN
increase_salary (90, .50);
ROLLBACK;
END;
/

- Procedure dijalankan untuk menaikkan gaji karyawan di department 90 sebesar 50% , Namun setelah itu dilakukan ROLLBACK

Query :

```
CREATE OR REPLACE PROCEDURE increase_salary (
    department_id_in   IN employees.department_id%TYPE,
    increase_pct_in    IN NUMBER
)
IS
BEGIN
    FOR employee_rec IN (
        SELECT employee_id
        FROM employees
        WHERE department_id = increase_salary.department_id_in
    )
    LOOP
        UPDATE employees emp
        SET emp.salary = emp.salary + emp.salary * increase_salary.increase_pc
        WHERE emp.employee_id = employee_rec.employee_id;

        DBMS_OUTPUT.PUT_LINE('Updated ' || SQL%ROWCOUNT);
    END LOOP;
END increase_salary;
/

select * from employees;
```

Output :

Script Output | Query Result

All Rows Fetched: 3 in 0.002 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	24000	(null)	(null)	90
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	17000	(null)	100	90
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	17000	(null)	100	90

3. Apa yang terjadi jika statement ROLLBACK dihapus pada sql code no 4.

Jawaban :

Analisis :

Jika perintah ROLLBACK dihapus, maka perubahan yang dilakukan oleh procedure tidak dibatalkan.

Query :

```
BEGIN
    increase_salary(90, 0.50);
END;
/
SELECT * FROM employees WHERE department_id = 90;
```

Output :

Script Output × | Query Result ×

SQL | All Rows Fetched: 3 in 0.003 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPART
1	100	Steven	King	SKING	515.123.4567	17-JUN-03	AD_PRES	36000	(null)	(null)	
2	101	Neena	Kochhar	NKOCHHAR	515.123.4568	21-SEP-05	AD_VP	25500	(null)	100	
3	102	Lex	De Haan	LDEHAAN	515.123.4569	13-JAN-01	AD_VP	25500	(null)	100	